THREE EQUIVALENT FORMS OF THE EXISTENCE OF AN IMPURITY, INCLUDING OVER THE HENSELIZATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The author's closing sentence at source line 4794, 'We omit the details.', is retained verbatim at the end of the counted proof; the factorization and the choice of points are inside the span.

Allowed standard prerequisites (editorial).

- The quasi-finite and henselization reductions for impurities. Same chapter flat.tex, labels lemma-quasi-finite-impurity-elementary (proof line 4784, claimed as S14-algebra-quasi-finite-impurity) and lemma-impurity-on-henselization (proof line 4786). The first gives that a quasi-finite impurity can be replaced by one over an elementary etale neighbourhood, and the second gives the corresponding statement starting from the henselization.

Source reference. Original lemma 05J2: sources/stacks-flat/pinned-original-source.tex, source lines 4744–4744. Navigation pointer for \ref{lemma-impurity-on-henselization} used inside the counted span (line 4786); the argument that consumes it is retained in the span.

Source reference. Original lemma 05J0: sources/stacks-flat/pinned-original-source.tex, source lines 4645–4645. Navigation pointer for \ref{lemma-quasi-finite-impurity-elementary} used inside the counted span (line 4784); the argument that consumes it is retained in the span.

Source reference. Original reference 05FW: sources/stacks-flat/pinned-original-source.tex, source lines 4473–4473. Navigation pointer for \ref{situation-pre-pure} used inside the counted span (line 4768); the argument that consumes it is retained in the span.

Lemma 0.1. *In Situation 05FW the following are equivalent*

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(1) *there exists an impurity* $(S^h \to S, s' \rightsquigarrow s, \xi)$ *of* $\mathcal{F}$ *above* s *where* S^h *is the henselization of* S *at* s*,*
(2) *there exists an impurity* $(T \to S, t' \rightsquigarrow t, \xi)$ *of* $\mathcal{F}$ *above* s *such that* $(T, t) \to (S, s)$ *is an elementary étale neighbourhood, and*
(3) *there exists an impurity* $(T \to S, t' \rightsquigarrow t, \xi)$ *of* $\mathcal{F}$ *above* s *such that* $T \to S$ *is quasi-finite at* t*.*

Proof. As an étale morphism is locally quasi-finite it is clear that (2) implies (3). We have seen that (3) implies (2) in Lemma 05J0. We have seen that (1) implies (2) in Lemma 05J2. Finally, if $(T \to S, t' \rightsquigarrow t, \xi)$ is an impurity of $\mathcal{F}$ above s such that $(T, t) \to (S, s)$ is an elementary étale neighbourhood, then we can choose a factorization $S^h \to T \to S$ of the structure morphism $S^h \to S$. Choose any point $s' \in S^h$ mapping to t' and choose any $\xi' \in X_{s'}$ mapping to $\xi \in X_{t'}$. Then $(S^h \to S, s' \rightsquigarrow s, \xi')$ is an impurity of $\mathcal{F}$ above s. We omit the details. □